

Birla Institute of Technology & Science, Pilani
Work Integrated Learning Programmes Division
First Semester 2025-2026
Comprehensive Examination

EC-3 Makeup Examination

Course No. : AIMLCZC416
Course Title : Mathematical Foundations for Machine Learning
Nature of Exam : Open Book
Weightage : 40%
Duration : 2 hrs 30 mts
Date of Exam : 08-03-2026

No. of Pages = 2

No of questions = 6

Note to Students:

1. Please follow all the *Instructions to Candidates* given on the cover page of the answer book.
2. All parts of a question should be answered consecutively. Each answer should start from a fresh page.
3. Assumptions made if any, should be stated clearly at the beginning of your answer.
4. Notation used here are as per the text book.

Q1. Consider the matrix $A = \begin{bmatrix} -2 & 0 & 0 & 0 \\ -1 & 3 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ -1 & -1 & 0 & 5 \end{bmatrix}$

(a) Find all the eigen values of A. Find the eigen space V corresponding to the largest eigen value of A. Also find a basis and dimension of V. [1 + 2 + 2 = 5M]

(b) Are the columns of A linearly independent. Justify your answer. [2M]

Q2. (a) Prove that the matrix $A = \begin{bmatrix} 3 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$ is a symmetric positive definite matrix. [2M]

(b) Using this matrix define an inner product on R^3 . [1M]

(c) Show that the vectors $\begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix}$ are not orthogonal with respect the inner product you defined on R^3 . [2M]

(d) Using the Gram–Schmidt orthonormalization process, construct an orthonormal set from the vectors $\begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix}$ with respect to the inner product which you defined in part (b) [3M]

3. Consider the loss function $L(w_1, w_2) = w_1^4 + w_2^4 - (w_1 + w_2)$. Suppose we want to minimize this function using the Gradient Descent algorithm with initial point $(w_1^0, w_2^0) = (0, 0)$. Define the line search function $\phi(\alpha) = L((w_1^0, w_2^0) + \alpha d)$ where $d = -\nabla L(0, 0)$.

(a) Apply the binary search method on the interval $[0, 2]$ to determine the learning rate. At each iteration, compare the function values $\phi(m)$ and $\phi(m + 10^{-3})$, where m is the midpoint of the current interval. Compute the reduced interval after two iterations. [4M]

(b) Using the second derivative test, determine the minimum of $\phi(\alpha)$ and verify whether the reduced interval obtained in part (a) is shrinking toward the optimal value. [3M]

Q4. Consider the function $J(w_0, w_1) = (w_0 - 3)^2 + (w_1 - 3)^2$.

We wish to minimize $J(w_0, w_1)$ using the Gradient Descent algorithm with momentum. The initial parameter values are $(w_0^0, w_1^0) = (2, 2)$, the learning rate is $\eta = 0.05$, the momentum parameter is $\beta = 0.5$, and the initial velocity vector is $(v_0^0, v_1^0) = (0, 0)$. Compute the parameter value (w_0^1, w_1^1) after the first iteration and (w_0^2, w_1^2) after the second iteration. [4M]

Q5. Consider the dataset

$$X = \begin{bmatrix} 0 & 3 \\ 1 & 5 \\ 2 & 7 \end{bmatrix},$$

where each row represents an observation in $\mathbb{R}^2$ and the columns represent features.

1. Compute the covariance matrix S of the dataset X . [2M]
2. Determine the eigenvalues and eigenvectors of S . [3M]
3. Using Principal Component Analysis (PCA), identify the one-dimensional principal subspace of $\mathbb{R}^2$ onto which the data should be projected. [2M]

Q6. Consider the following training dataset in $\mathbb{R}^2$:

Point	x_1	x_2	Class y
X_1	3	3	+1
X_2	5	5	+1
X_3	3	0	-1
X_4	5	0	-1

(a) Let $w = (w_1, w_2)^T \in \mathbb{R}^2$ be the weight vector, $b \in \mathbb{R}$ be the bias term, $\xi_i \geq 0$, $i = 1, 2, 3, 4$, be the slack variables, and $C > 0$ be the regularization parameter. Write the primal optimization problem for the soft-margin Support Vector Machine for the given dataset. [1M]

(b) Show that the soft-margin SVM problem can be written in the unconstrained form:

$$\min_{w, b} \frac{1}{2} \|w\|^2 + C \sum_{i=1}^4 \max(0, 1 - y_i(w^T X_i + b)). \quad [2M]$$

(c) Suppose $w = (1, -1)^T$ and $b = 0$.

1. Compute $y_i(w^T X_i + b)$ for each data point. [2M]
2. Compute the hinge loss for each point. [1M]

3. Determine the total objective value for $C = 1$. [1M]
